



ROHDE & SCHWARZ

Unternehmensbereich
Rundfunk- und Fernsehtechnik

Circuit Description

TV Test Receiver

EMF ... Power Supply

821.4519

1 Power Supply

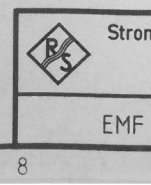
See 822.0917 S

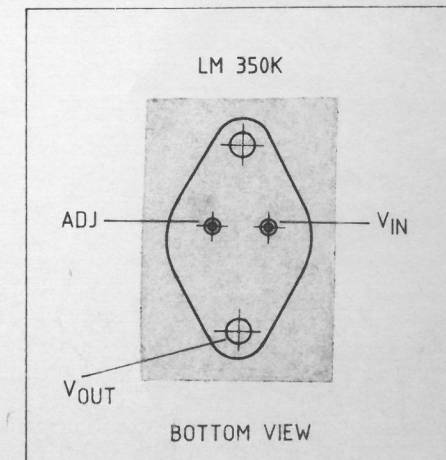
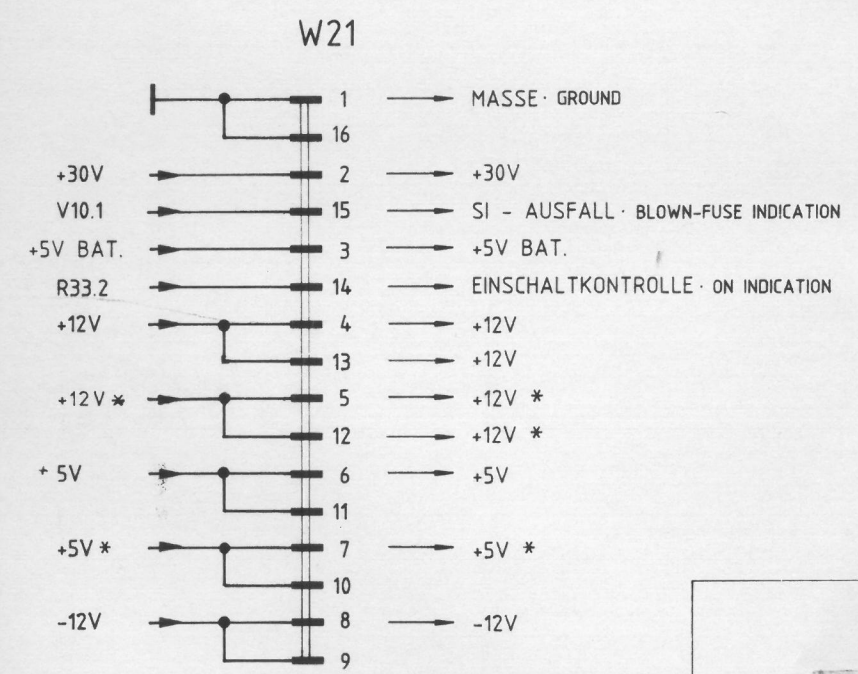
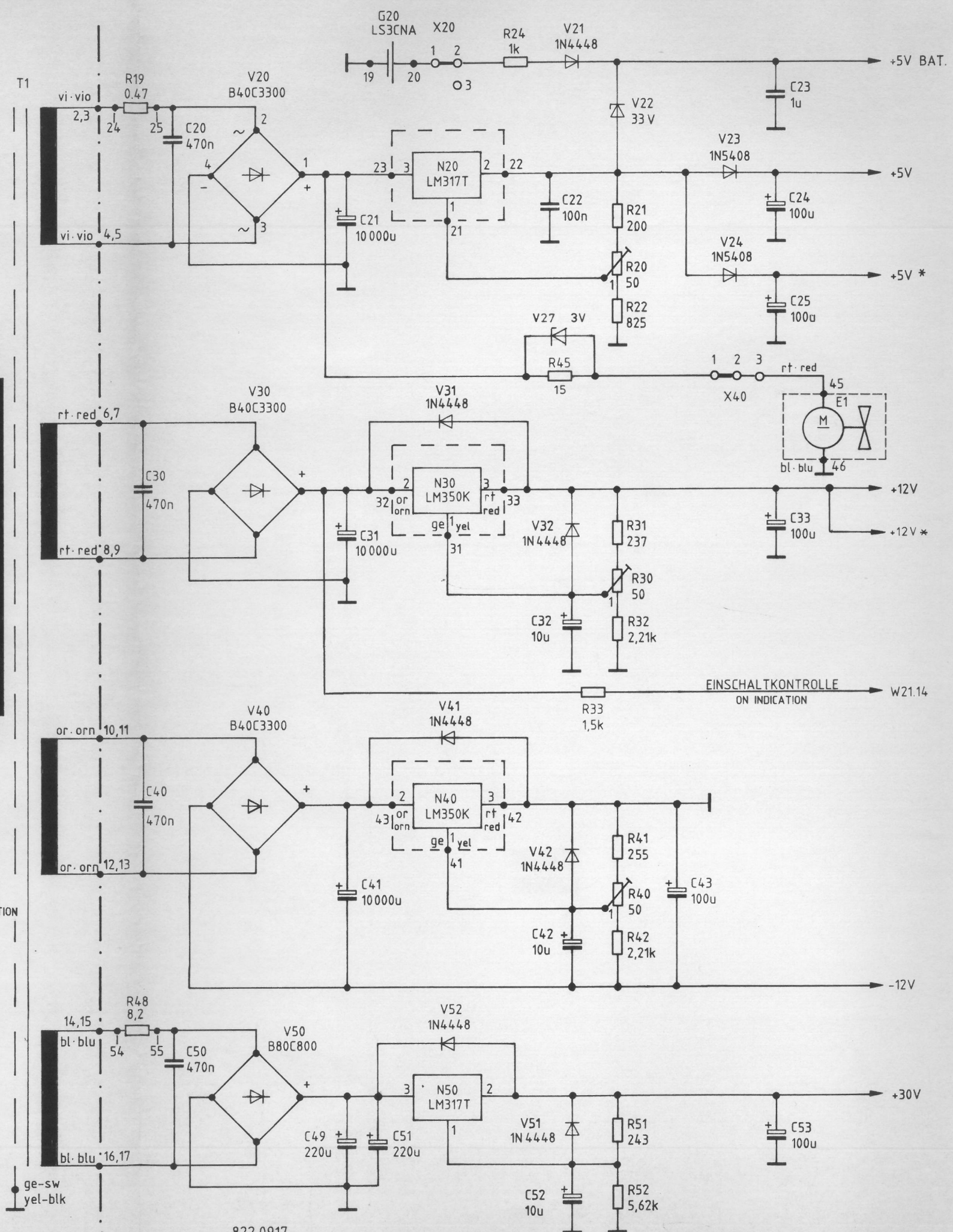
The power supply is located at the rear of the instrument as a compact assembly. It supplies regulated voltages of + 30 V, + 12 V, -12 V and + 5 V.

The AC supply voltage passes through an RFI suppression filter, is switched by S1 and applied to the AC transformer T1 via fuse F1, a thermal fuse and the voltage selector. The voltage selector is integrated in subassembly Z1 together with the Euro connector and the RFI suppression filter and fitted at the rear of the instrument. The AC supply is applied to transformer T10 if the fuse blows. The secondary voltage of T10 is rectified and is used for the message "Blown fuse". A generously dimensioned bridge rectifier and smoothing filter in conjunction with the integrated voltage regulators ensure that a reliable power supply is provided. All voltages apart from + 30 V are adjustable.

The lithium battery G20 is used to store a selected channel and important operating states and operates via V21/V22. The voltage regulators are fitted on a generously dimensioned heat sink. The built-in blower provides additional operational reliability.

2 KGH	gezeichnet	bearbeitet	geprüft	normgepr.
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	Stromlauf zu		NETZTEILPLATTE		Zeichn.-Nr.	Blatt-Nr.
			POWER SUPPLY BOARD			
EMF	reg. i. V.	821.4019 V	erste Z.	821.4519 S		
			v. 1 Bl.			

A



C

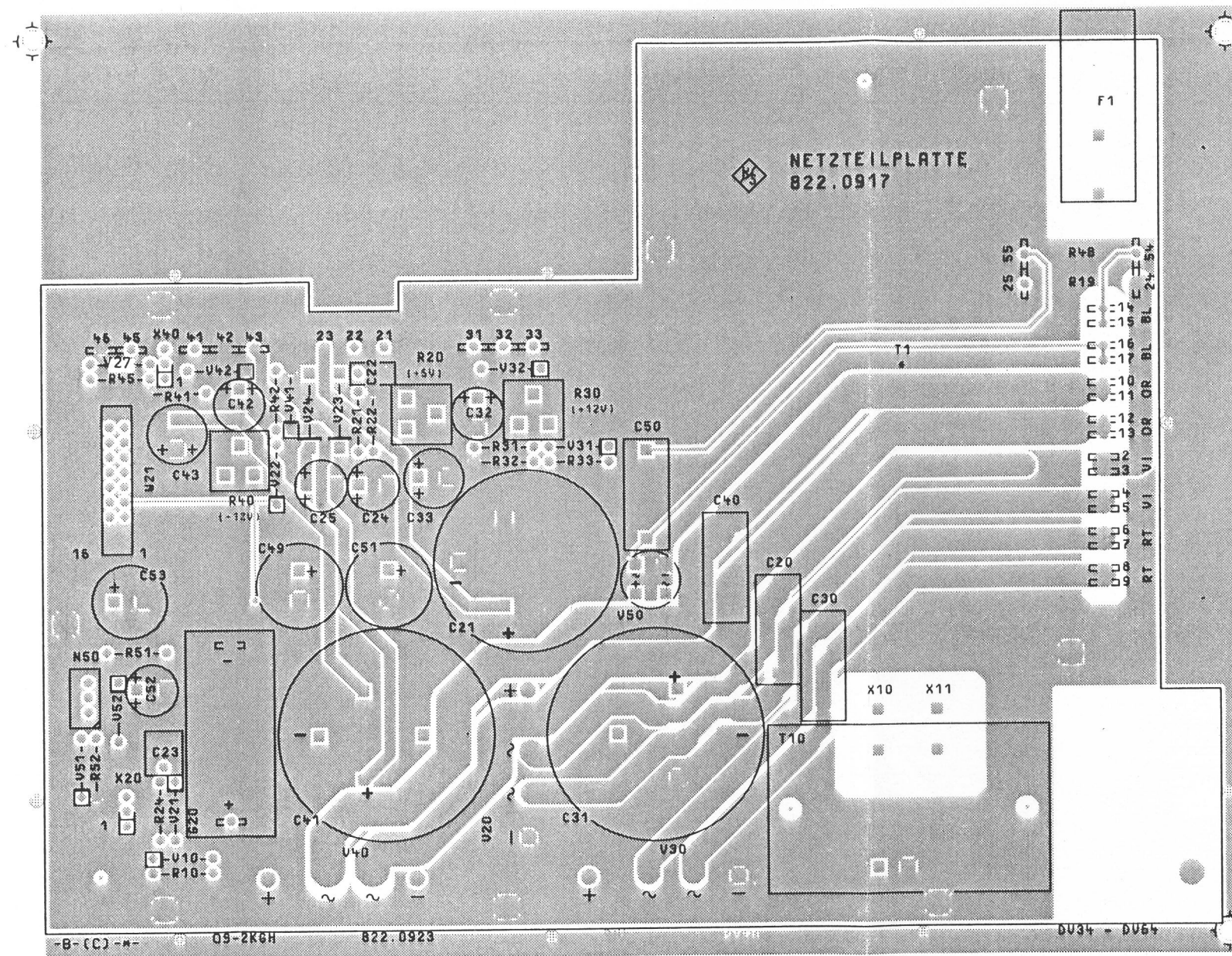
D

E

F

B		07.87	OS	Maße ohne Toleranzangabe		Maßstab 1 : 1					
C	40163	23.02.88	OS			Halbzeug, Werkstoff					
D	40163	05.89	OS								
				2KGH	Tag	Name	Benennung NETZTEILPLATTE POWER - SUPPLY - BOARD				Z
				Bearb.	07.87	OS					
				Gepr.							
				Norm							
				 ROHDE & SCHWARZ			Zeichn.-Nr.		Blatt-Nr.		
And. Zust.	Anderungs- Mitteilung	Tag	Name				822.0917.01		ED		3
				zu Gerät EMF		reg. i. V. 821.4019 V		erste Z.		v Bl	

Ansicht und Leitungsführung Bauteilseite
View of tracks on component side



VARIANTENERKLÄRUNG / VERSION
VAR 02 - GRUNDAUSFÜHRUNG / BASIC MODEL

B	C	40163	07.87	OS	Maße ohne Toleranzangabe	Maßstab 1 : 1	Halbzeug, Werkstoff	Benennung	Z
					2KGH	Tag	Name	NETZTEILPLATTE POWER - SUPPLY - BOARD	
					Bearb.	07.87	OS		
					Gepr.				
					Norm			Zeichn.-Nr	Blatt-Nr
								822.0917.01	ED
								reg. i. V. 821.4019 V	erste Z.
Änd Zust	Änderungs- Mitteilung	Tag	Name		zu Gerät	EMF			

(hierzu HVC 250)



ACHTUNG: EGB!
Elektrostatisch gefährdete
Bauelemente erfordern eine
besondere Handhabung
ATTENTION ESD!
Electrostatic sensitive
devices require a special
handling.

ROHDE&SCHWARZ		AZ	Datum Date	Schaltteilliste für Parts list for	Sachnummer Stock Nr.	Blatt Page
		05	0787	ZE NETZTEIL	821.4519.01 SA	1
Kennzeichen Component No.	Benennung/Beschreibung Designation	Sachnummer Stock No.	enthalten in contained in			
C20	CK 470NF+-20%100VQUADER PLASTIC-FOIL CAPACITOR ROEDERST MKT1822-447/0	CK 006.5079	822.0917.01			
C21	CE 10000UF-10+50%16V30X50 ELECTROLYTIC CAPACITOR ROEDERST EYV00BB510D	CE 219.3459	822.0917.01			
C22	CK 100NF+-5%63V5RM MKT CAPACITOR WIMA MKS/2/63/0,1UF/5%	CK 099.2930	822.0917.01			
C23	CK 1UF+-10%50V5RM MKT CAPACITOR WIMA MKS2/50/1UF/10%	CK 099.2998	822.0917.01			
C24	CE 100UF-10+50% 16V 9X13 ELECTROLYTIC CAPACITOR ROEDERST EK 00CB 310 D	CE 006.7165	822.0917.01			
C25	CE 100UF-10+50% 16V 9X13 ELECTROLYTIC CAPACITOR ROEDERST EK 00CB 310 D	CE 006.7165	822.0917.01			
C30	CK 470NF+-20%100VQUADER PLASTIC-FOIL CAPACITOR ROEDERST MKT1822-447/0	CK 006.5079	822.0917.01			
C31	CE 10000UF10+50%40V35X 80 ELECTROLYTIC CAPACITOR ROEDERST ELKOEYV10000/40ISO	CE 250.3134	822.0917.01			
C32	CE 10UF -10+50% 63V 9X13 ELECTROLYTIC CAPACITOR ROEDERST ELKOEK10/63	CE 022.7650	822.0917.01			
C33	CE 100UF-10+50% 25V 13X13 ELECTROLYTIC CAPACITOR ROEDERST ELKOEK100/25	CE 208.4007	822.0917.01			
C40	CK 470NF+-20%100VQUADER PLASTIC-FOIL CAPACITOR ROEDERST MKT1822-447/0	CK 006.5079	822.0917.01			
C41	CE 10000UF10+50%40V35X 80 ELECTROLYTIC CAPACITOR ROEDERST ELKOEYV10000/40ISO	CE 250.3134	822.0917.01			
C42	CE 10UF -10+50% 63V 9X13 ELECTROLYTIC CAPACITOR ROEDERST ELKOEK10/63	CE 022.7650	822.0917.01			
C43	CE 100UF-10+50% 25V 13X13 ELECTROLYTIC CAPACITOR ROEDERST ELKOEK100/25	CE 208.4007	822.0917.01			
C49	CE 220UF-10+50% 63V 15X30 ELECTROLYTIC CAPACITOR SIEMENS ELKO B4136-A8227-Z	CE 086.4316	822.0917.01			
C50	CK 470NF+-20%100VQUADER PLASTIC-FOIL CAPACITOR ROEDERST MKT1822-447/0	CK 006.5079	822.0917.01			
C51	CE 220UF-10+50% 63V 15X30 ELECTROLYTIC CAPACITOR SIEMENS ELKO B4136-A8227-Z	CE 086.4316	822.0917.01			
C52	CE 10UF -10+50% 63V 9X13 ELECTROLYTIC CAPACITOR ROEDERST ELKOEK10/63	CE 022.7650	822.0917.01			

821.4519.01 SA BL 1+

ROHDE&SCHWARZ		AZ	Datum Date	Schaltteilleiste für Parts list for	Sachnummer Stock Nr.	Blatt Page
		05	0787	ZE NETZTEIL	821.4519.01 SA	2
Kennzeichen Component No.	Benennung/Beschreibung Designation	Sachnummer Stock No.	enthalten in contained in			
C53	CE 100UF-10+50% 63V 15X20 ELECTROLYTIC CAPACITOR SIEMENS ELKOB 41316-A8107-Z	CE 086.4300	822.0917.01			
EI	EV 80X80X38 17L/S 12V- BLOWER 12V PAPST 8112G AXIALLUEFTER	801.8160				
F1	SS SCHMELZS.T1 DIN41662 FUSE WICKMANN T1 DIN 41662 TROP FUER 220V SS T2 DIN41571 SS020.7546 FOR 100V	SS 020.7446	822.0917.01			
G20	EB 3,4V LITHIUM-BATTERIE LI BATTERY SAFT LS 3 CNA	565.1687	822.0917.01			
N20	BO LM317T +ADJ1A5 VREGL VOLTAGE REGULATOR NSC LM317T	BO 339.4080				
N30	BO LM350K +ADJ3A0 VREGL ADJUSTABLE VOLT.REGULATOR NSC LM350K	664.2045				
N40	BO LM350K +ADJ3A0 VREGL ADJUSTABLE VOLT.REGULATOR NSC LM350K	664.2045				
N50	BO LM317T +ADJ1A5 VREGL VOLTAGE REGULATOR NSC LM317T	BO 339.4080	822.0917.01			
R10	RL 0,35W 1KOHM+-1%TK50 RESISTOR DRALORIC SMA0207/1K-F-C	RL 082.2160	822.0917.01			
R19	RD 0,8W 0,47 OHM+-3% WIRE-WOUND RESISTOR SAGE 1000S/0,47OHM/3%	RD 069.1464	822.0917.01			
R20	RS 0,5W50 OHM+-10%10X10X5 CERMET POTENTIOMETER T BOURNS 3386F-1-500	RS 247.7861	822.0917.01			
R21	RL 0,35W 200 OHM+-1%TK50 RESISTOR DRALORIC SMA0207/200OHM-F-D	RL 083.0049	822.0917.01			
R22	RL 0,35W 825 OHM+-1%TK50 RESISTOR DRALORIC SMA 0207/825OHM-F-C	RL 082.2502	822.0917.01			
R24	RL 0,35W 1KOHM+-1%TK50 RESISTOR DRALORIC SMA0207/1K-F-C	RL 082.2160	822.0917.01			
R30	RS 0,5W50 OHM+-10%10X10X5 CERMET POTENTIOMETER T BOURNS 3386F-1-500	RS 247.7861	822.0917.01			
R31	RL 0,35W 237 OHM+-1%TK50 RESISTOR DRALORIC SMA0207/237OHM-F-D	RL 083.0110	822.0917.01			
821.4519.01 SA BL 2+						

ROHDE&SCHWARZ		AZ	Datum Date	Schaltteilliste für Parts list for	Sachnummer Stock Nr.	Blatt Page
		05	0787	ZE NETZTEIL	821.4519.01 SA	3
Kennzeichen Component No.	Benennung/Beschreibung Designation	Sachnummer Stock No.	enthalten in contained in			
R32	RL 0,35W 2,21KOHM+-1%TK50 RESISTOR	RL 082.2477	822.0917.01			
R33	DRALORIC SMA 0207/2,21K-F-C RL 0,35W 1,50KOHM+-1%TK50 RESISTOR	RL 083.0732	822.0917.01			
R40	DRALORIC SMA0207/1,50K-F-D RS 0,5W50 OHM+-10%10X10X5 CERMET POTENTIOMETER T	RS 247.7861	822.0917.01			
R41	BOURNS 3386F-1-500 RL 0,35W 255 OHM+-1%TK50 RESISTOR	RL 083.0149	822.0917.01			
R42	DRALORIC SMA0207/255OHM-F-D RL 0,35W 2,21KOHM+-1%TK50 RESISTOR	RL 082.2477	822.0917.01			
R44	DRALORIC SMA 0207/2,21K-F-C RL 0,35W 10,0 OHM+-1%TK50 RESISTOR	RL 082.8852	822.0917.01			
R45	DRALORIC SMA0207/10OHM-F-D RL 0,35W 10,0 OHM+-1%TK50 RESISTOR	RL 082.8852	822.0917.01			
R48	DRALORIC SMA0207/10OHM-F-D RL 8,2 OHM 10% SCHUTZWID. PROTECTIVE RESISTOR	558.8215	822.0917.01			
R51	RESISTA NKS2 8,2 OHM 10% RL 0,35W 243 OHM+-1%TK50 DEPOS.-CARBON RESISTOR	RL 083.0126	822.0917.01			
R52	DRALORIC SMA0207/243OHM-F-D RL 0,35W 5,62KOHM+-1%TK50 RESISTOR	RL 082.2190	822.0917.01			
	DRALORIC SMA0207/5,62K-F-C					
S1	SB NETZSCHALTER 2XU O.KN. POWER SWITCH ITT SF OKN NE18 2 U E E	SB 007.5143				
T1	LT RINGKERNTRAFO	821.8743				
T10	LT 220V/12V 1VA GS TRANSFORMER ISMET MMG954/51	248.7794	822.0917.01			
V10	AD 1N4448 75V 0,15A UDI DIODE	AD 012.0700	822.0917.01			
V20	TEXAS INST 1N4448 GEGURTET AG B40C5000/3300 BRGL RECTIFIER	AG 084.5115	822.0917.01			
V21	SIEMENS B40C5000/3300SI AD 1N4448 75V 0,15A UDI DIODE	AD 012.0700	822.0917.01			
V22	TEXAS INST 1N4448 GEGURTET AE BZX79/C33 0,5W Z-DI ZENER DIODE	AE 012.2632	822.0917.01			
V23	VALVO BZX79/C33 AG 1N5804 GL 100V 2A5 RECTIFIER UNITRODE 1N5804	AG 453.4762	822.0917.01			
821.4519.01 SA BL 3+						

821.4519.01 SA BL 3+

ROHDE&SCHWARZ		AZ	Datum Date	Schaltteilliste für Parts list for	Sachnummer Stock Nr.	Blatt Page
		05	0787	ZE NETZTEIL	821.4519.01 SA	4
Kennzeichen Component No.	Benennung/Beschreibung Designation	Sachnummer Stock No.	enthalten in contained in			
V24	AG 1N5804 GL 100V 2A5 RECTIFIER	AG 453.4762	822.0917.01			
V30	UNITRODE 1N5804 AG B40C5000/3300 BRGL RECTIFIER	AG 084.5115	822.0917.01			
V31	SIEMENS B40C5000/3300SI AD 1N4448 75V 0,15A UDI DIODE	AD 012.0700	822.0917.01			
V32	TEXAS INST 1N4448 GEGURTET AD 1N4448 75V 0,15A UDI DIODE	AD 012.0700	822.0917.01			
V40	TEXAS INST 1N4448 GEGURTET AG B40C5000/3300 BRGL RECTIFIER	AG 084.5115	822.0917.01			
V41	SIEMENS B40C5000/3300SI AD 1N4448 75V 0,15A UDI DIODE	AD 012.0700	822.0917.01			
V42	TEXAS INST 1N4448 GEGURTET AD 1N4448 75V 0,15A UDI DIODE	AD 012.0700	822.0917.01			
V50	TEXAS INST 1N4448 GEGURTET AG B80C800 BRGL RECTIFIER	AG 013.2042	822.0917.01			
V51	AEG-TELEF. B80C800SI AD 1N4448 75V 0,15A UDI DIODE	AD 012.0700	822.0917.01			
V52	TEXAS INST 1N4448 GEGURTET AD 1N4448 75V 0,15A UDI DIODE	AD 012.0700	822.0917.01			
	TEXAS INST 1N4448 GEGURTET					
W21	DX BANDKABEL	821.8714	822.0917.01			
X10	FV FLACHSTECKER 2,8X0,8 FLAT PLUG 2,8X0,8 VOGT 3775A/0,8/MS-S18	FV 279.1998	822.0917.01			
X11	FV FLACHSTECKER 2,8X0,8 FLAT PLUG 2,8X0,8 VOGT 3775A/0,8/MS-S18	FV 279.1998	822.0917.01			
X20	FP INDIREKT.STECKERL.36P. PIN CONNECTOR BERG 75160-102-36 1X2-POLIG	FP 242.3600	822.0917.01			
X40	FP INDIREKT.STECKERL.36P. PIN CONNECTOR BERG 75160-102-36 1X3-POLIG	FP 242.3600	822.0917.01			
Z1	FN NETZFILT.M.SPANNUNGSW. FILTER SCHAFFNER FN369-2/01	FN 099.3313				
- ENDE -						
821.4519.01 SA BL 4-						